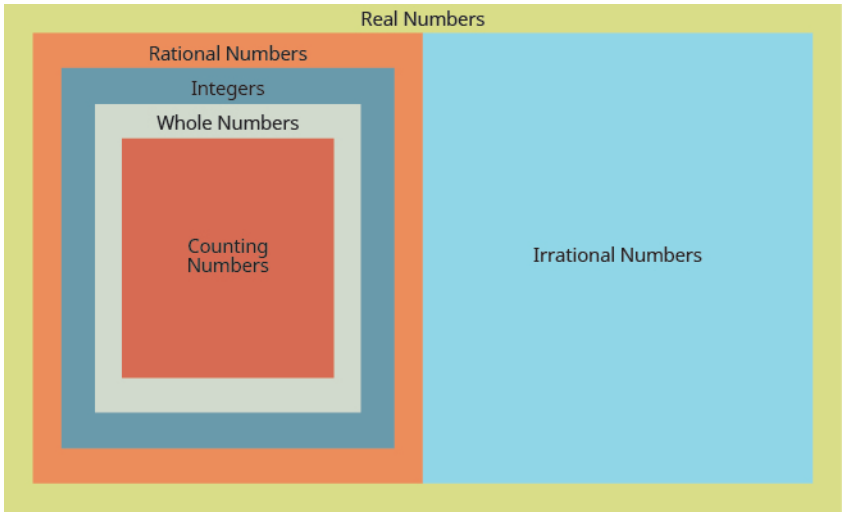
Chapter Review

Key Terms

Additive Identity The **additive identity** is 0. When zero is added to any number, it does not change the value.
Additive Inverse The opposite of a number is its **additive inverse**. The **additive inverse** of a is $-a$.
Irrational number An **irrational number** is a number that cannot be written as the ratio of two integers. Its decimal form does not stop and does not repeat.
Multiplicative Identity The **multiplicative identity** is 1. When one multiplies any number, it does not change the value.
Multiplicative Inverse The reciprocal of a number is its **multiplicative inverse**. The **multiplicative inverse** of a is $\frac{1}{a}$.
Rational number A **rational number** is a number that can be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$. Its decimal form stops or repeats.
Real number a **real number** is a number that is either rational or irrational.

Key Concepts

7.1 Rational and Irrational Numbers

- **Real numbers**
  - 

7.2 Commutative and Associative Properties

- **Commutative Properties**
  - **Commutative Property of Addition:**
    - If a, b are real numbers, then $a + b = b + a$
  - **Commutative Property of Multiplication:**
    - If a, b are real numbers, then $a \cdot b = b \cdot a$
- **Associative Properties**
  - **Associative Property of Addition:**
    - If a, b, c are real numbers then $(a + b) + c = a + (b + c)$
  - **Associative Property of Multiplication:**
    - If a, b, c are real numbers then $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

7.3 Distributive Property

- **Distributive Property:**
  - If a, b, c are real numbers then
    - $a(b + c) = ab + ac$
    - $(b + c)a = ba + ca$
    - $a(b - c) = ab - ac$

7.4 Properties of Identity, Inverses, and Zero

- **Identity Properties**
  - **Identity Property of Addition:** For any real number a: $a + 0 = a \quad 0 + a = a$ **0** is the **additive identity**
  - **Identity Property of Multiplication:** For any real number a: $a \cdot 1 = a \quad 1 \cdot a = a$ **1** is the **multiplicative**